3. Project Planning Phase

3.1 Introduction

The project planning phase establishes the objectives, scope, resources, activities, schedule, and risks of LegalEase. Its purpose is to create a clear roadmap so that development activities are organized, resources are used effectively, and the project is completed within the specified time.

3.2 Project Objectives

- To develop an easy-to-use AI-powered legal document generator.
- To provide secure user registration and authentication.
- To integrate Google Gemini models for drafting, explanation, and review.
- To collect accurate details through a guided questionnaire.
- To make legal text understandable through plain-language explanations.
- To allow editing, version saving, and export as PDF or Word.
- To store documents securely in a personal history.

3.3 Project Scope

The scope includes the design and development of a web-based legal drafting assistant. Users can create an account, log in, choose a document type, enter details, generate a draft, edit clauses, request explanations and reviews, export the document, and view saved documents. Electronic signatures, lawyer consultation, court filing, automatic legal updates, and mobile apps are outside the current scope and are listed as future enhancements. The output is a draft for information and is not legal advice.

3.4 Project Modules

- Authentication Module
- Document Selection and Input Module
- AI Document Generator Module
- Clause Explanation Module
- Review and Risk Analysis Module
- Editor and Version Module
- Export Module
- Document History Module

3.5 Resource Planning

Resources include the development computer, internet connection, Python and FastAPI, a web frontend, a relational database, a Google Gemini API key, ReportLab and python-docx for export, Visual Studio Code, and Git/GitHub for version control.

3.6 Project Schedule

Phase	Activities	Duration
1	Brainstorming, problem identification, idea finalization	Week 1
2	Requirement analysis and documentation	Week 1-2

Phase	Activities	Duration
3	System design, architecture, database design	Week 2-3
4	Backend and frontend development, Gemini integration	Week 3-6
5	Testing and bug fixing	Week 6-7
6	Demonstration preparation and final documentation	Week 7-8

3.7 Risk Analysis

Risk	Impact	Mitigation
Gemini API downtime or rate limits	Drafting temporarily unavailable	Error handling, retry logic, user-friendly messages
Incorrect or inappropriate legal wording	User may rely on a flawed draft	Structured prompts, review feature, and a visible disclaimer advising review by a qualified lawyer
Differences between jurisdictions	Draft may not suit local law	Jurisdiction field included in every prompt; disclaimer
Invented details or clauses by AI	Incorrect document	Prompt rules to use only supplied facts and use placeholders for missing data
Misuse for unlawful documents	Legal and ethical risk	Input validation, Gemini safety settings, blocked-request messages
Leakage of personal data	Privacy risk	Authentication, per-user data filtering, no document sharing
Malformed AI output	Document cannot be displayed	Strict output format, parsing with validation and fallback handling
API key exposure	Security and misuse risk	Store key in server environment variables
Schedule delays	Incomplete features	Prioritize core modules first